Supplemental Materials

Impact of vaccination on carriage and infection by antibiotic-resistant bacteria: a systematic review and meta-analysis

This document contains an additional set of information for the paper titled: "Impact of vaccination on carriage and infection by antibiotic-resistant bacteria: a systematic review and meta-analysis". More specifically, this Supplement 1 contains:

Table S1. Additional information on the papers included in the meta-analysis.

Fig. S1. Forest plot for effect of PCV on the incidence of serotype-targeted infections nonsusceptible to antibiotic, by type of antibiotic category.

Fig. S2. Forest plot for effect of PCV on the incidence of all pneumococcal infections.

Fig. S3. Forest plot for effect of PCV on the incidence of infections nonsusceptible to antibiotic, by age of participants.

Fig. S4. Forest plot for effect of PCV on the incidence of infections nonsusceptible to antibiotic, by type of vaccine.

Fig. S5. Forest plot for effect of PCV on the incidence of infections nonsusceptible to antibiotic, by population group.

Fig. S6. Additional analyses and sensitivity analyses, bacterial carriage (prevalence).

Fig. S7. Forest plot for effect of PCV on the carriage of serotype-targeted infections nonsusceptible to antibiotic, by type of antibiotic category.

Fig. S8. Forest plot for effect of PCV on the carriage of all pneumococcus infections.

Fig. S9. Forest plot for effect of PCV on the carriage of infections nonsusceptible to antibiotic by, age of participants.

Fig. S10. Forest plot for effect of PCV on the carriage of infections nonsusceptible to antibiotic, by type of vaccine.

Fig. S11. Forest plot for effect of PCV on the carriage of infections nonsusceptible to antibiotic, by population group.

Table S1. Additional information on the papers included in the meta-analysis

Author, year	Timeline of the study (years)	Source of the data	No. of records	Quality (grade)	Reasons for downgrading	Notes
Black et al. 2004 [20]	1998-1999, 2001-2002	Hospital records, Northern California Kaiser Permanente	37,793	Low	Non-RCT, unclear or high risk of bias related to blinding, limitations in study design	Coverage of the vaccination programme: 74%
Cho et al. 2012 [21]	1997-1998, 2001-2002, 2009-2010	Hospital records, Seoul National University Children's hospital	1,243	Low	Unclear or high risk of bias in multiple domains - confounding factors	Vaccination coverage increased from 14% to 72%
Cho et al. 2014 [22]	2006-2010	Hospital records from 8 hospitals	140	Very low	Unclear or high risk of bias in multiple domains, bias related to blinding	
Cohen et al. 2006 [23]	2001-2004	Ambulatory records from 89 paediatricians	1,906	Very low	High risk related to blinding, unclear or high risk of bias in multiple domains	Vaccination coverage increased from 8.2% to 61.4%
Diawara et al. 2015 [24]	2007- 2010, 2011-2014	Laboratory and hospital records	136	Very low	Non-RCT, unclear or high risk of bias relating to blinding, limitations in study design due to confounding variables	
Hammitt et al. 2006 [25]	1998-2004	Community-wide surveys	15,598	Low	Non-RCT, unclear or high risk of bias related to blinding, limitations in study design	
Hampton et al. 2012 [26]	1998- 2008	Active bacterial core surveillance	43,198	Very low	Non-RCT, limitations in study design due to confounding factors	
Hennessy et al. 2005 [27]	1995-2003	National immunization survey	1,113	Very low	Non-RCT, limitations in study design due to confounding factors	Vaccination coverage increased from 26% to 75%
Hsu et al. 2009 [28]	1998-2005	Active bacterial core surveillance through clinical microbiology laboratories	1,379	Very low	Non-RCT, limitations in data collection, limitations in study design due to confounding factors	
Kaplan et al. 2004 [29]	1994-2000, 2001-2002	Hospital records from 8 hospitals	270	Low	Unclear or high risk of bias in multiple domains - confounding factors	
Klugman et al. 2003 [30]	1998-2000	Hospital records	39,836	High	-	All individuals received *Haemophilus influenzae* type b vaccine
Kyaw et al. 2006 [31]	1996-2004	Active Bacterial Core Surveillance	28,336	Very low	Non-RCT, limitations in study design due to confounding factors	
Marom et al. 2017 [32]	2008-2009, 2010-2013	Ministry of Health and hospital records	295	Very low	Unclear or high risk related to blinding, small sample size, limitations in study design	
Mbelle et al. 1999 [33]	2008-2009, 2010-2013	Hospital records, Chris Hani Baragwanath Hospital	492	High	Unclear or high risk related to blinding, small sample size, limitations in study design	All individuals received Diphtheria-Tetanus toxoids-Pertussis, oral Polio, Hepatitis b, and *H. influenzae* type b vaccine; study including a population with high prevalence of HIV
Sigurdsson et al. 2017 [34]	2009-2015	Data from day care centres	147	Very low	Non RCT, lack of blinding, limitations in study design due to confounding factors	
Stephens et al. 2005 [35]	1999-2002	Active bacterial core surveillance, Georgia Health District 3	5,589	Very low	Non RCT, lack of blinding, limitations in study design due to confounding factors	
Tomczyk et al. 2016 [36]	2009-2013	Active bacterial core surveillance	20,270	Low	Non-RCT, limitations in study design resulting in confounding factors	
von Gottberg et al. 2014 [37]	2005-2008, 2011-2012	Hospital records	35,192	Low	Non-RCT, unclear or high risk of bias related to blinding, limitations in study design	Study including a population with high prevalence of HIV
Whitney et al. 2003 [38]	1998-2001	Active bacterial core surveillance	13,568	Low	Non-RCT, unclear or high risk of bias related to blinding, limitations in study design	

Numbers in squared brackets reported in the column author; year refers to the number of the reference as in the article; time line cells reporting two intervals refer to periods used in before-after studies.
RCT, randomized controlled trial; HIV, human immunodeficiency virus.

 https://doi.org/10.7774/cevr.2021.10.2.81

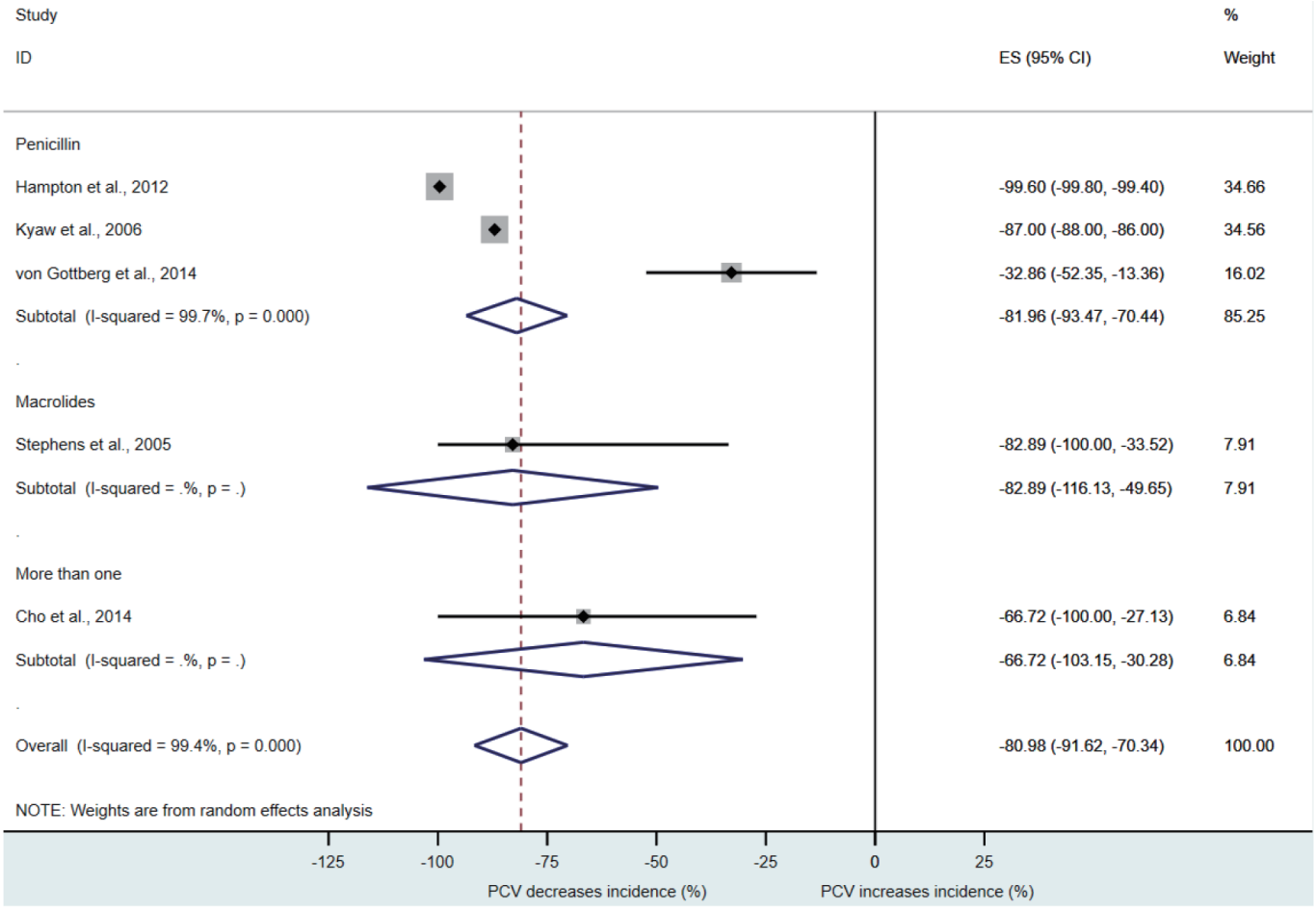


Fig. S1. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the incidence of serotype-targeted infections nonsusceptible to antibiotic, by type of antibiotic category. ES, estimates; CI, confidence interval.

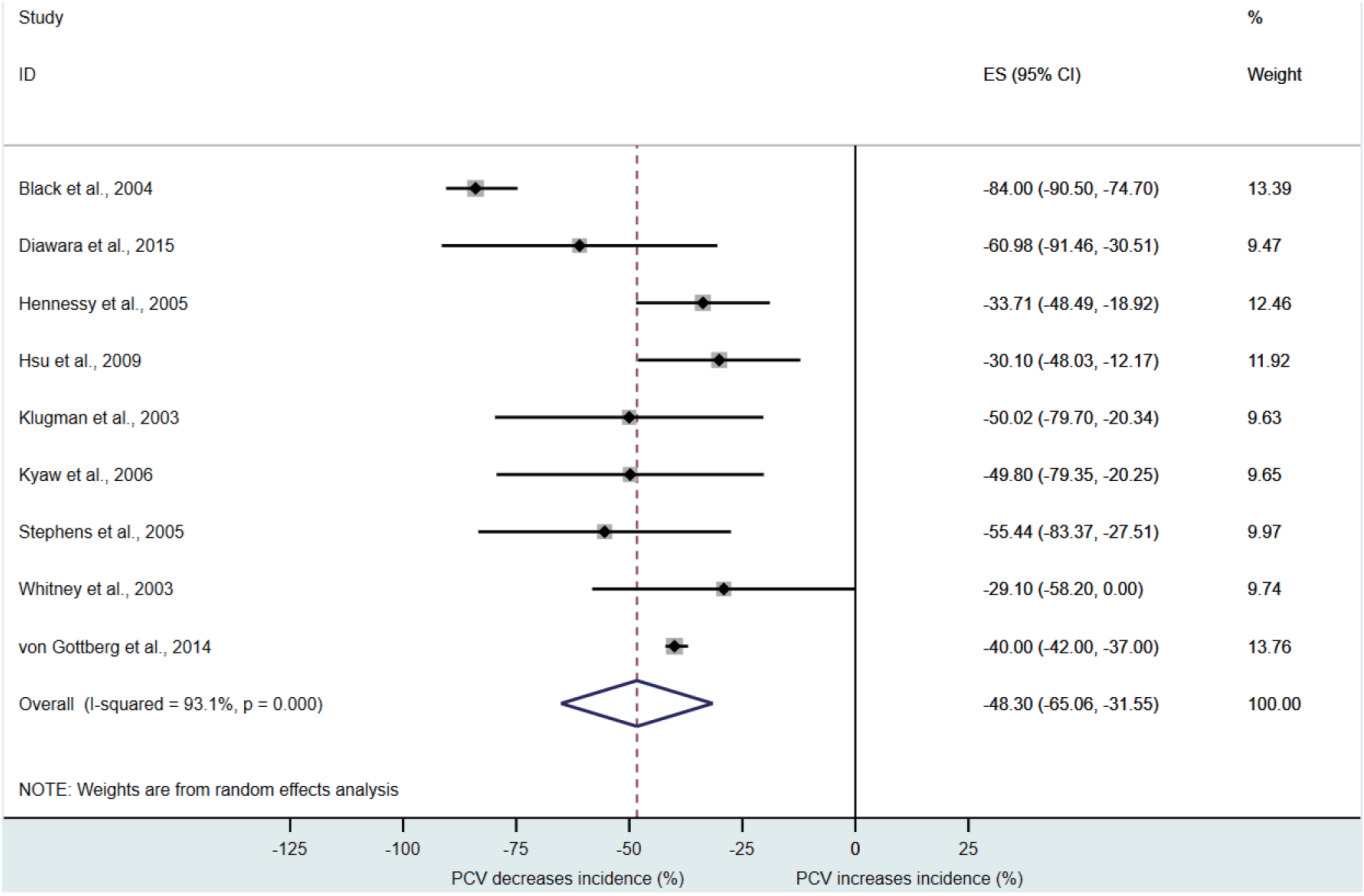


Fig. S2. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the incidence of all pneumococcal infections. ES, estimates; CI, confidence interval.

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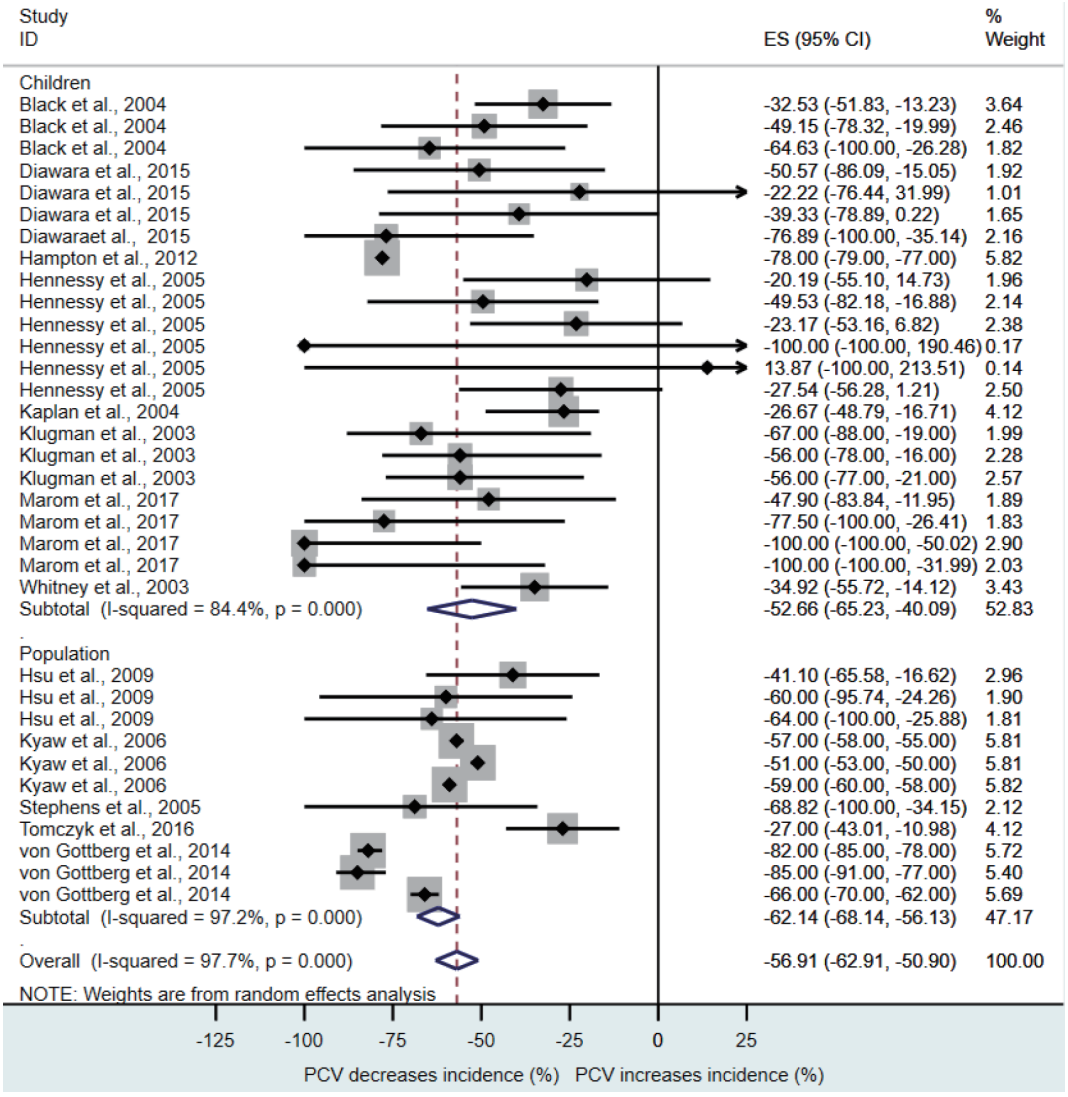


Fig. S3. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the incidence of infections nonsusceptible to antibiotic, by age of participants. ES, estimates; CI, confidence interval.

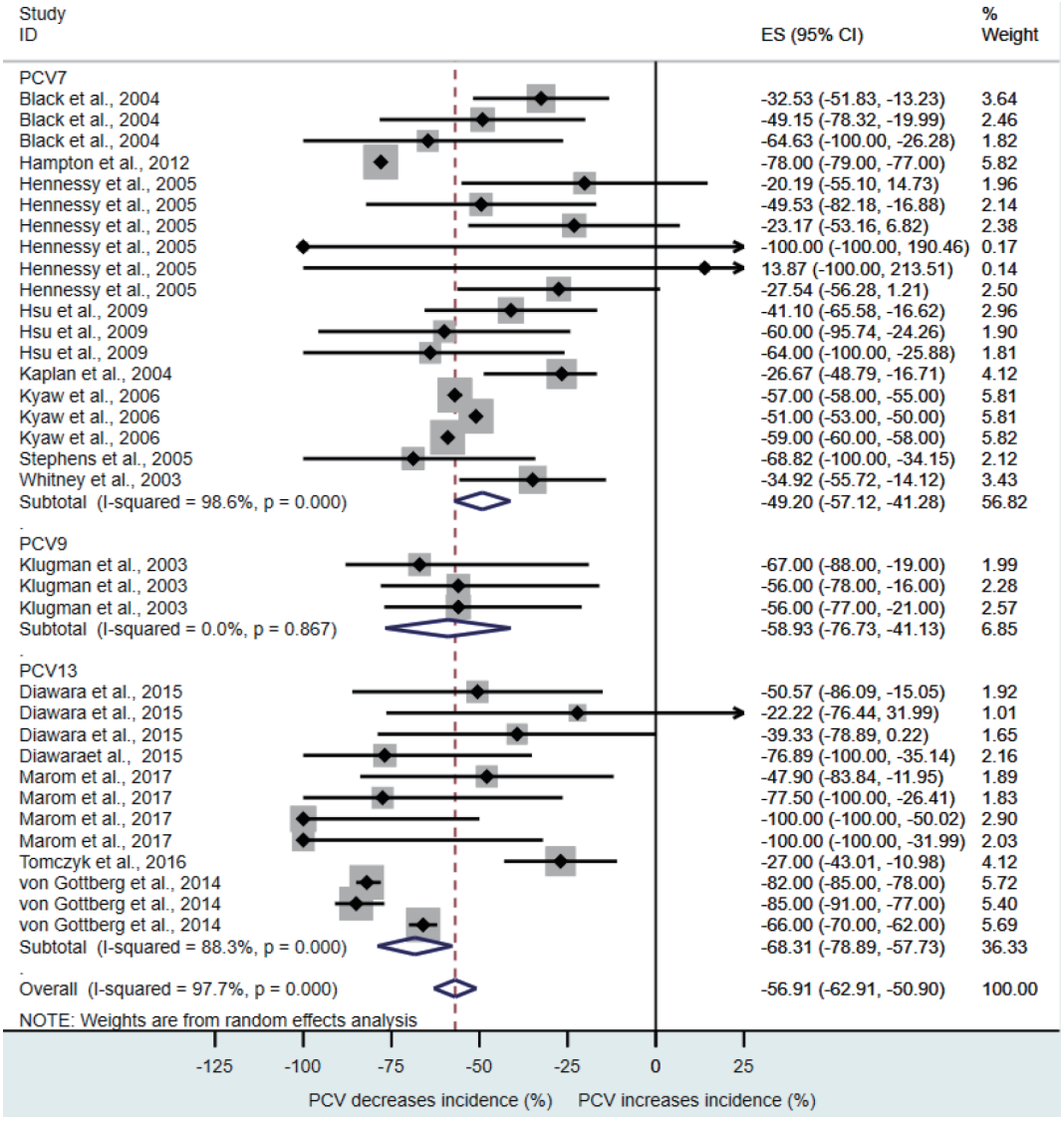


Fig. S4. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the incidence of infections nonsusceptible to antibiotic, by type of vaccine. ES, estimates; CI, confidence interval.

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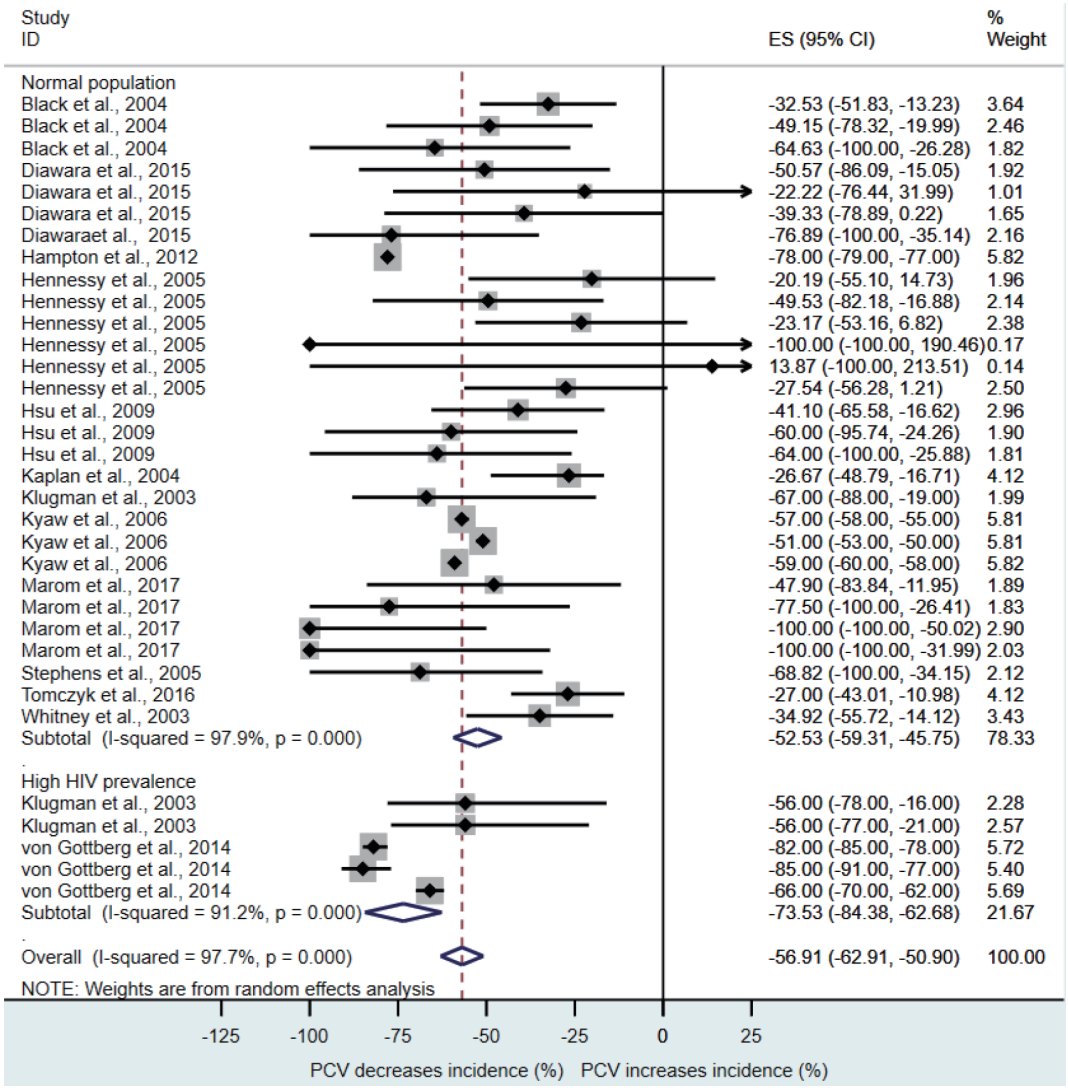


Fig. S5. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the incidence of infections nonsusceptible to antibiotic, by population group. ES, estimates; CI, confidence interval.

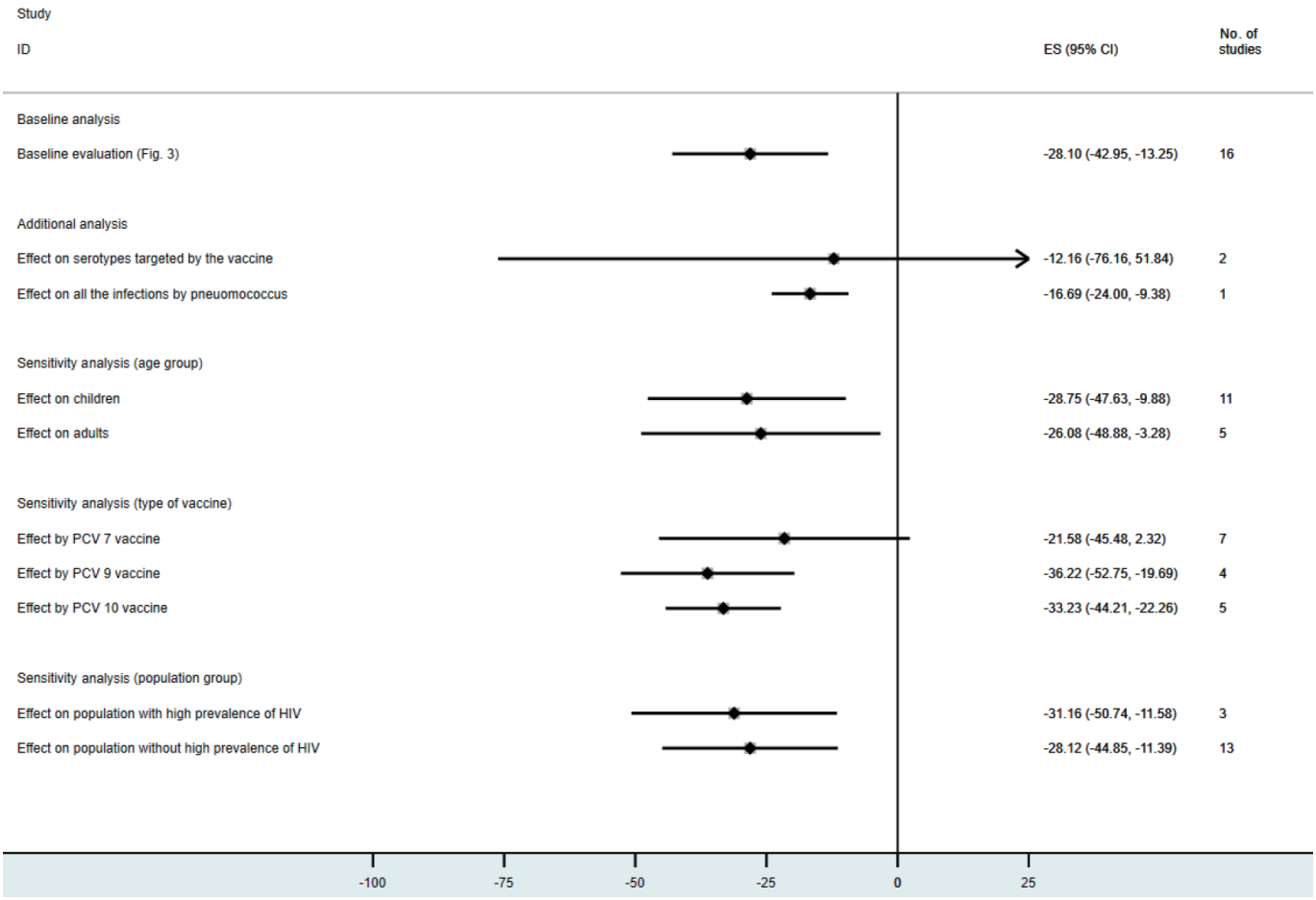


Fig. S6. Additional analyses and sensitivity analyses, bacterial carriage (prevalence). ES, estimates; CI, confidence interval; PCV, pneumococcal conjugate vaccines; HIV, human immunodeficiency virus.

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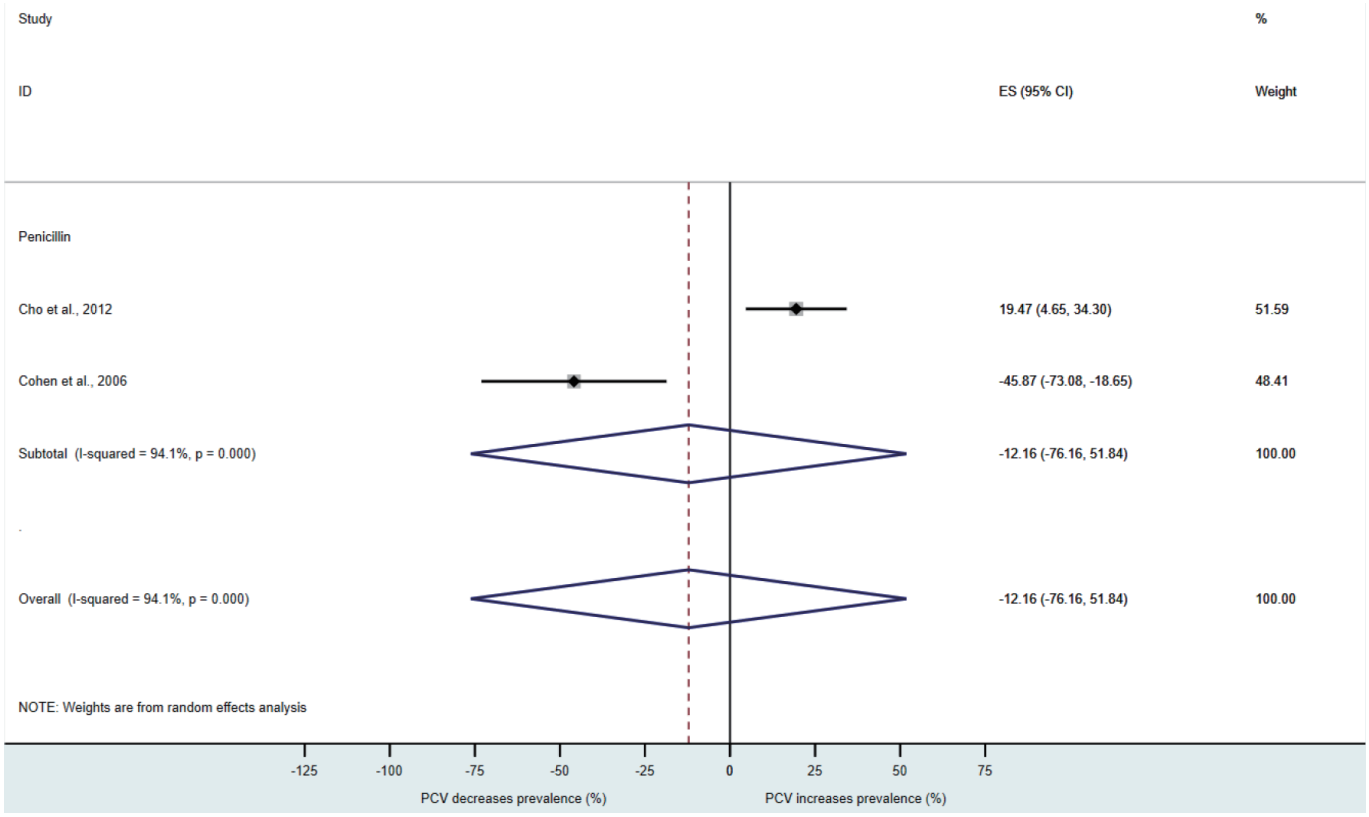


Fig. S7. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the carriage of serotype-targeted infections nonsusceptible to antibiotic, by type of antibiotic category. ES, estimates; CI, confidence interval.

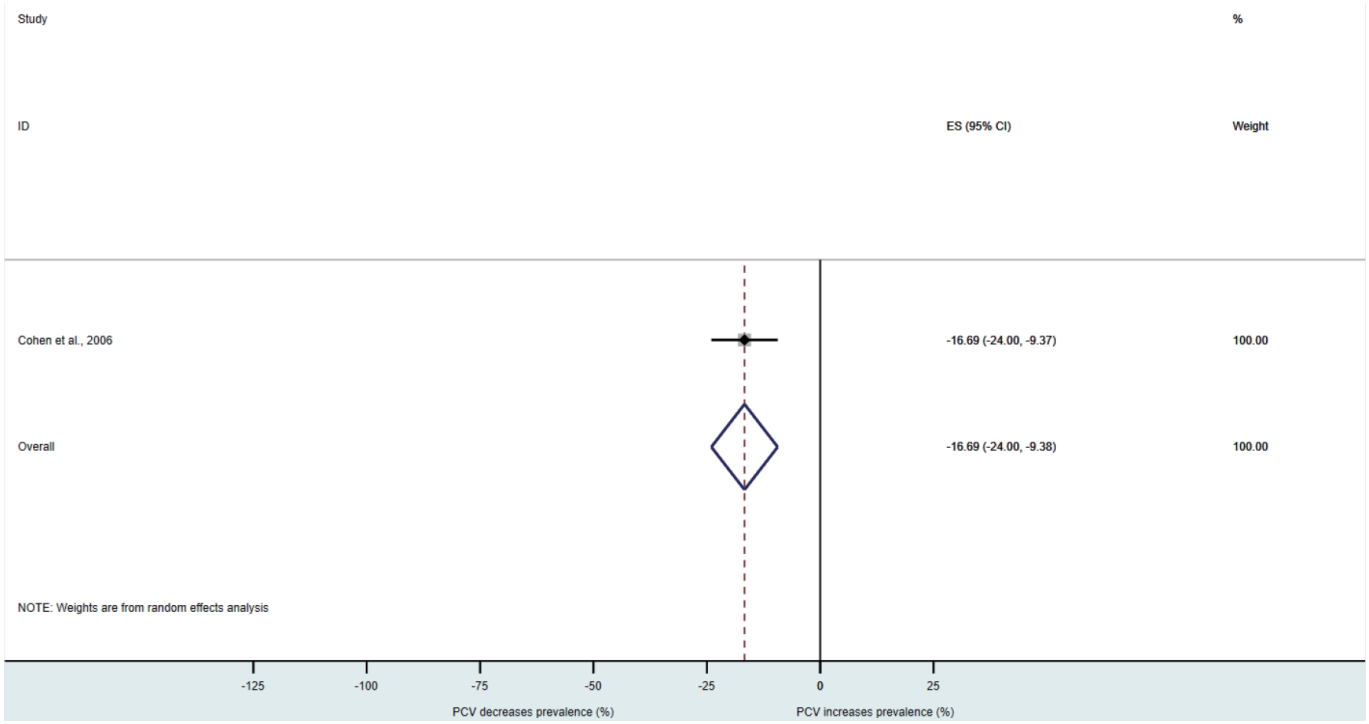


Fig. S8. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the carriage of all pneumococcus infections. ES, estimates; CI, confidence interval.

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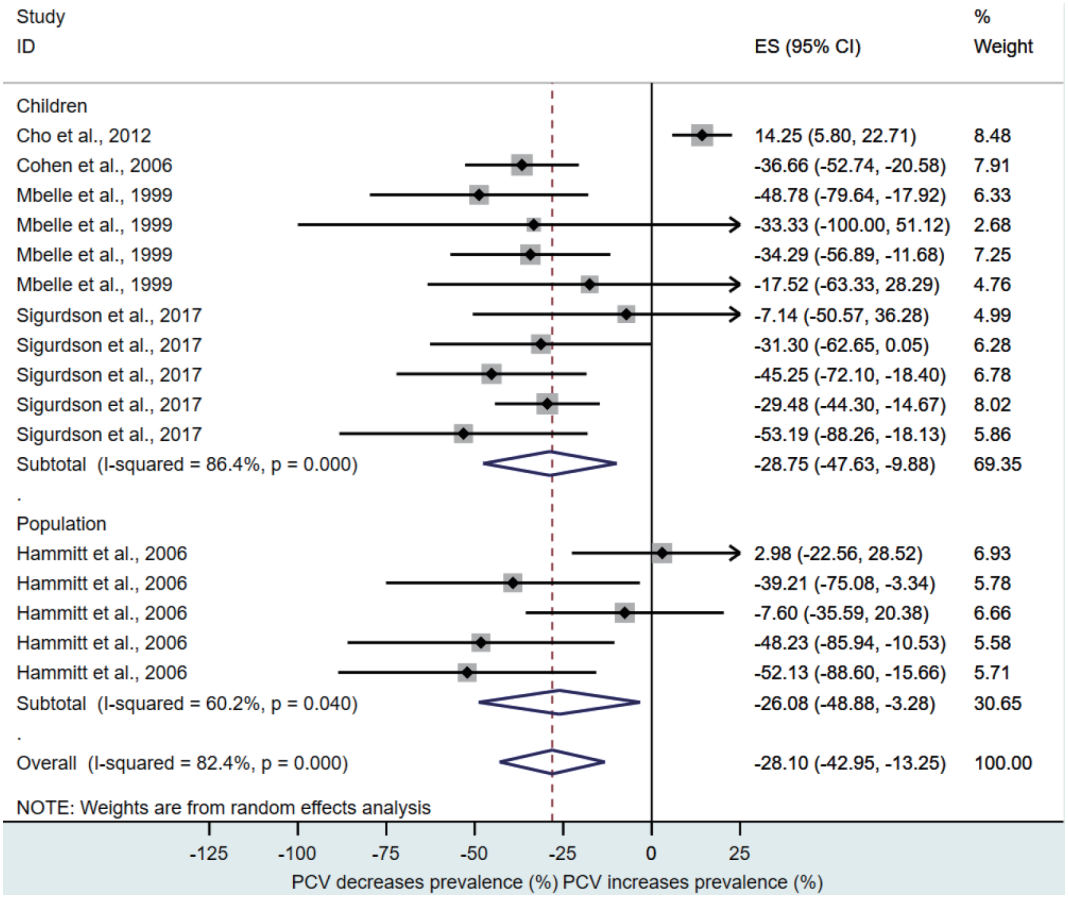


Fig. S9. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the carriage of infections nonsusceptible to antibiotic by, age of participants. ES, estimates; CI, confidence interval.

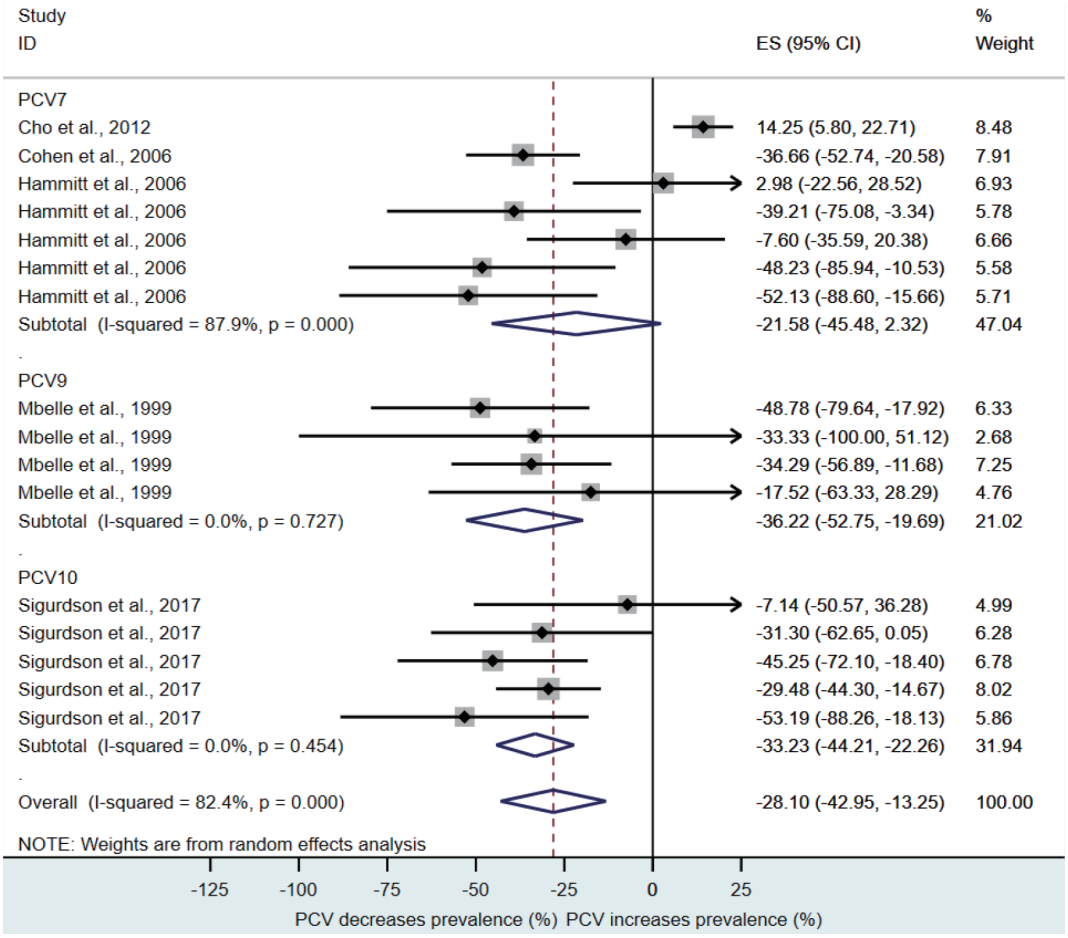


Fig. S10. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the carriage of infections nonsusceptible to antibiotic, by type of vaccine. ES, estimates; CI, confidence interval.

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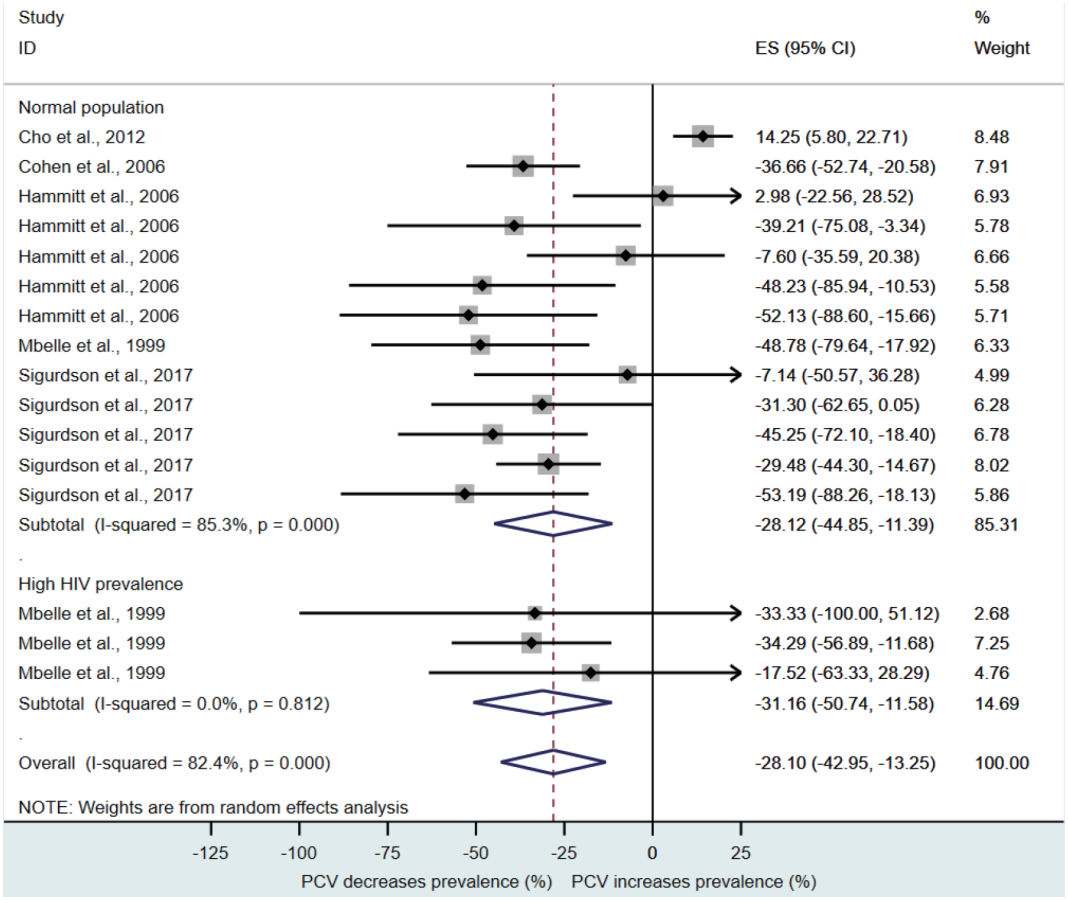


Fig. S11. Forest plot for effect of pneumococcal conjugate vaccine (PCV) on the carriage of infections nonsusceptible to antibiotic, by population group. ES, estimates; CI, confidence interval.